

Optimal Transport: Intuition

Optimal transport studies how to move one distribution of mass into another in the cheapest possible way.

Think of a pile of sand and a hole with the same total volume but different shapes. Moving sand costs something depending on how far it travels. The question is: what is the cheapest way to move all the sand to fill the hole?

In general, we replace sand with a probability distribution μ and the hole with another distribution ν . Both have total mass 1.

Monge formulation

We look for a map

$$T : X \rightarrow Y$$

that sends each point x to a single point $T(x)$ such that μ is transformed into ν . The cost is

$$\int_X c(x, T(x)) d\mu(x).$$

Problem: this requires each x to go to exactly one y , which may not be possible.

Kantorovich formulation

Instead of moving each point directly, we allow mass splitting. We use a transport plan $\pi(x, y)$, which tells us how much mass moves from x to y . We then minimize total cost:

$$\int_{X \times Y} c(x, y) d\pi(x, y).$$

This always has a solution and is more flexible than Monge's formulation.

Duality

Instead of thinking about moving mass, we can think about pricing.

Suppose:

- $\phi(x)$ = cost of taking mass from x
- $\psi(y)$ = cost of delivering mass to y

To avoid losing money, these must satisfy:

$$\phi(x) + \psi(y) \leq c(x, y).$$

The dual problem is:

$$\max \left(\int \phi d\mu + \int \psi d\nu \right).$$

Key result: the optimal transport cost equals the best possible pricing strategy.

Regularized Optimal Transport

Exact optimal transport is expensive to compute, so we modify the problem slightly.

We add entropy regularization:

$$\min_{\pi \in \Pi(\mu, \nu)} \langle C, \pi \rangle - \epsilon H(\pi),$$

where entropy encourages smoother (less extreme) transport plans.

This has two important effects:

- The solution becomes unique
- The problem becomes much easier to solve numerically

As $\epsilon \rightarrow 0$, we recover the original optimal transport solution.

Sinkhorn form

The solution has the structure:

$$\pi_\epsilon^* = \text{diag}(u) K \text{diag}(v), \quad K = e^{-C/\epsilon}.$$

We compute u and v by iterating:

$$u \leftarrow \frac{\mu}{Kv}, \quad v \leftarrow \frac{\nu}{K^T u}.$$

This is fast and scalable.

Colour Transfer

Optimal transport can be used to transfer the “style” of colours from one image to another.

Each pixel is represented as a point in RGB space, so an image becomes a distribution of colours.

We compute an optimal transport plan between source and target colour distributions. Since a pixel may map to multiple colours, we convert the result into a single colour using a barycentric projection (a weighted average).

Practical issues

Two main problems arise:

1. Noise: nearby pixels may map to very different colours. This can be reduced using entropy regularization or by including spatial information so nearby pixels stay similar.

2. Computational cost: images can have hundreds of thousands of pixels. To handle this:

- We sample a subset of pixels
- Run Sinkhorn on the reduced problem
- Assign remaining pixels based on nearest samples

We may also limit iterations of Sinkhorn to speed up computation, trading accuracy for efficiency.